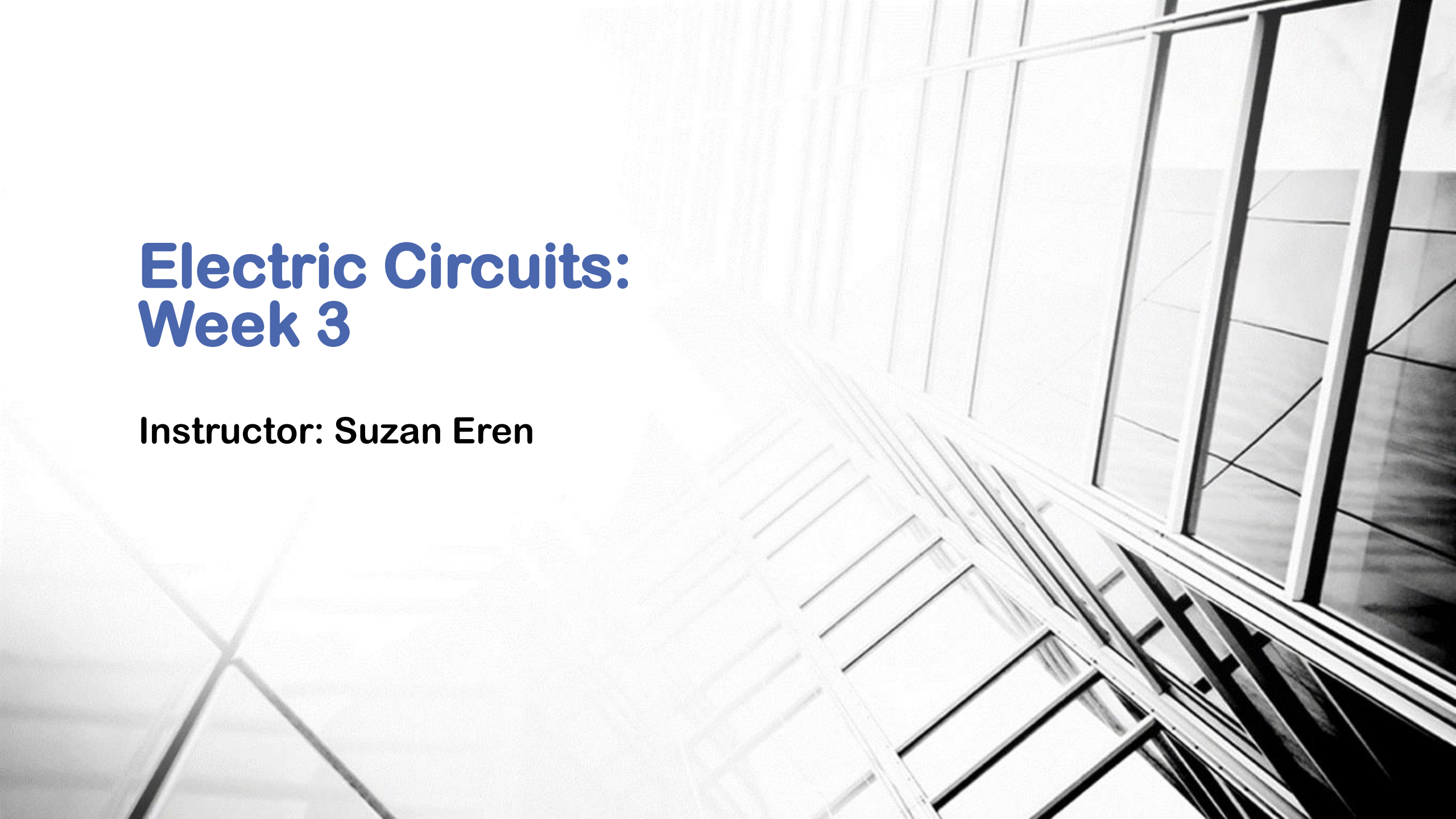
Electric Circuits: Week 3

Instructor: Suzan Eren



New Circuit Analysis Techniques

- So far, we've learned how to solve simple circuits. How do we solve more structurally complicated circuits?

- We're going to learn some new techniques that will help us analyze/solve circuits:

(1) ***Node-Voltage Method***

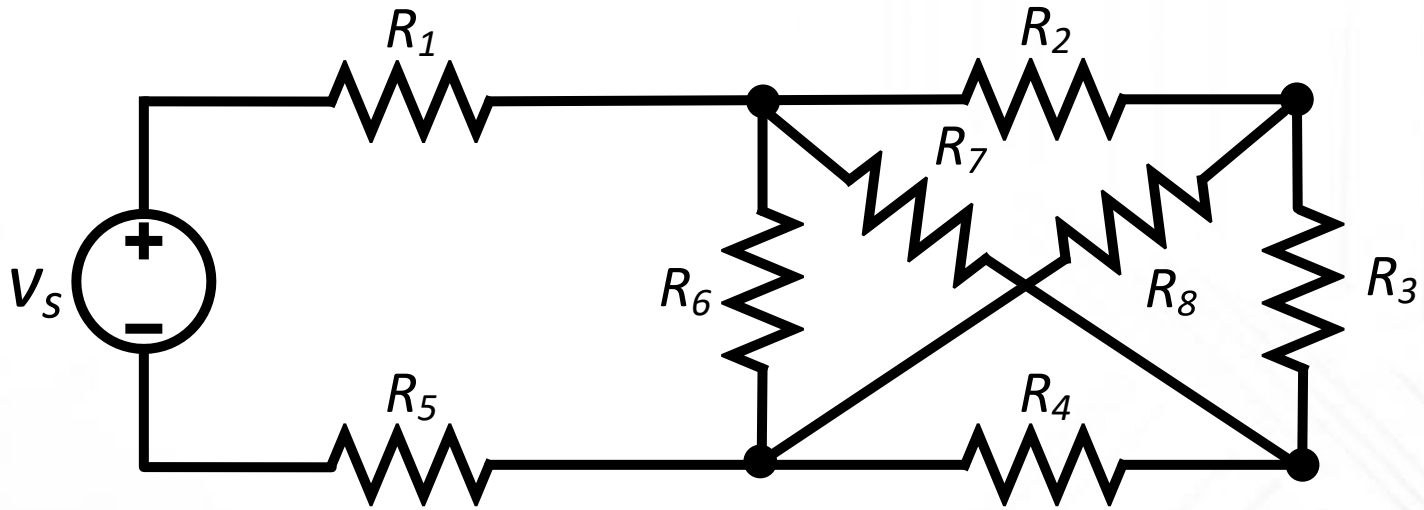
(2) ***Mesh-Current Method***

…Let's start with some terminology!

Terminology: Planar Circuit

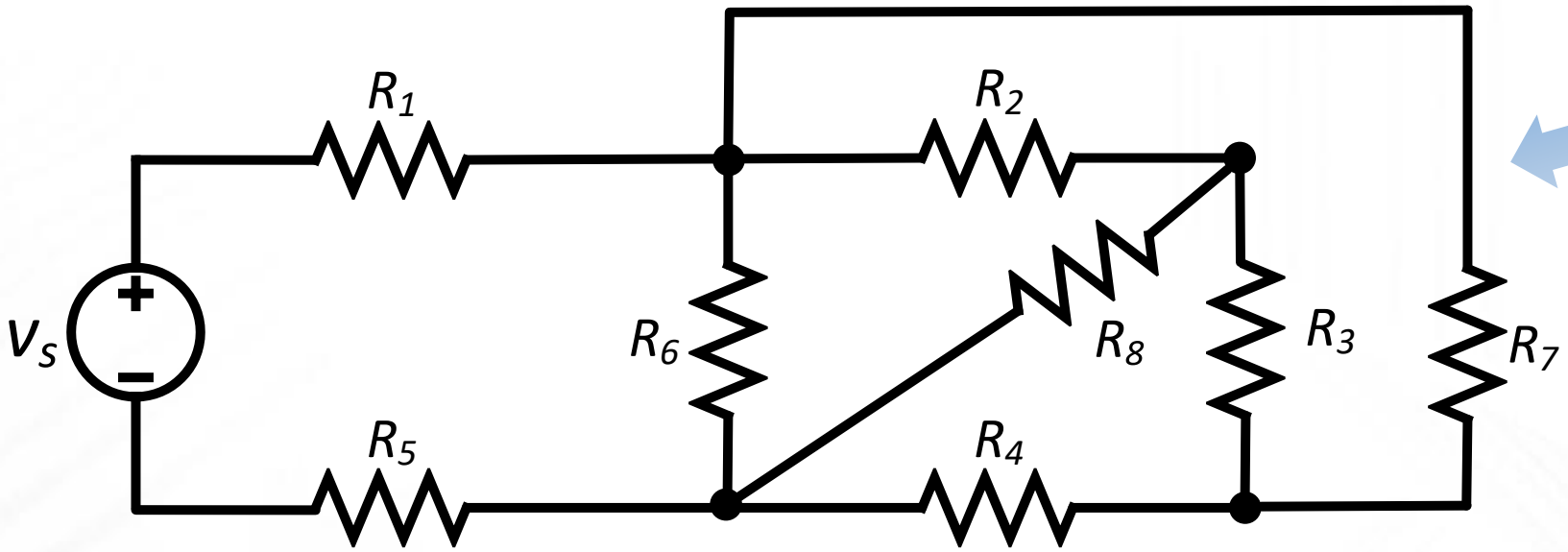
Planar Circuit: A circuit that can be drawn on a plane with no crossing branches

Example: Is this circuit planar?



Terminology: Planar Circuit

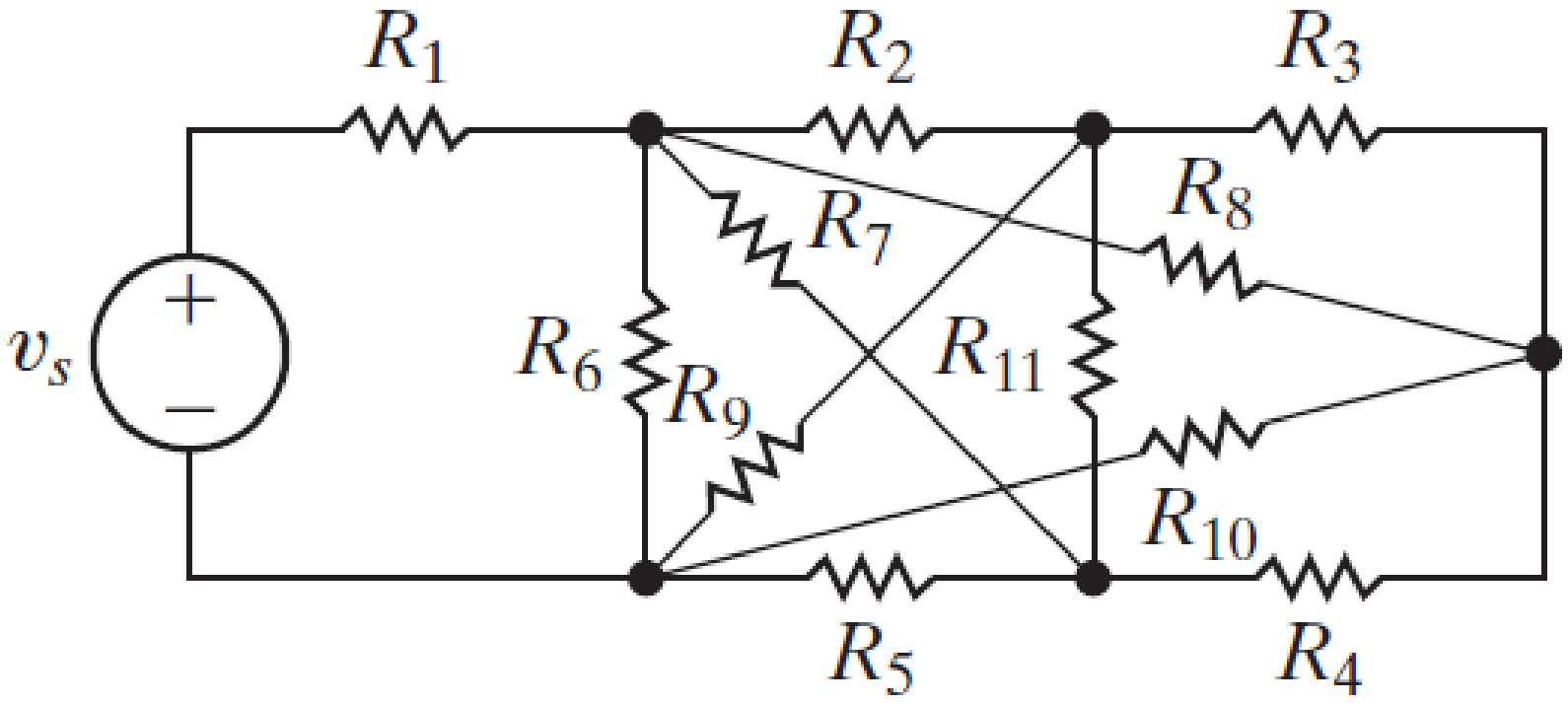
Yes! Because it can be redrawn as:



No crossing branches!

Terminology: Planar Circuit

An example of a **non-planar** circuit:

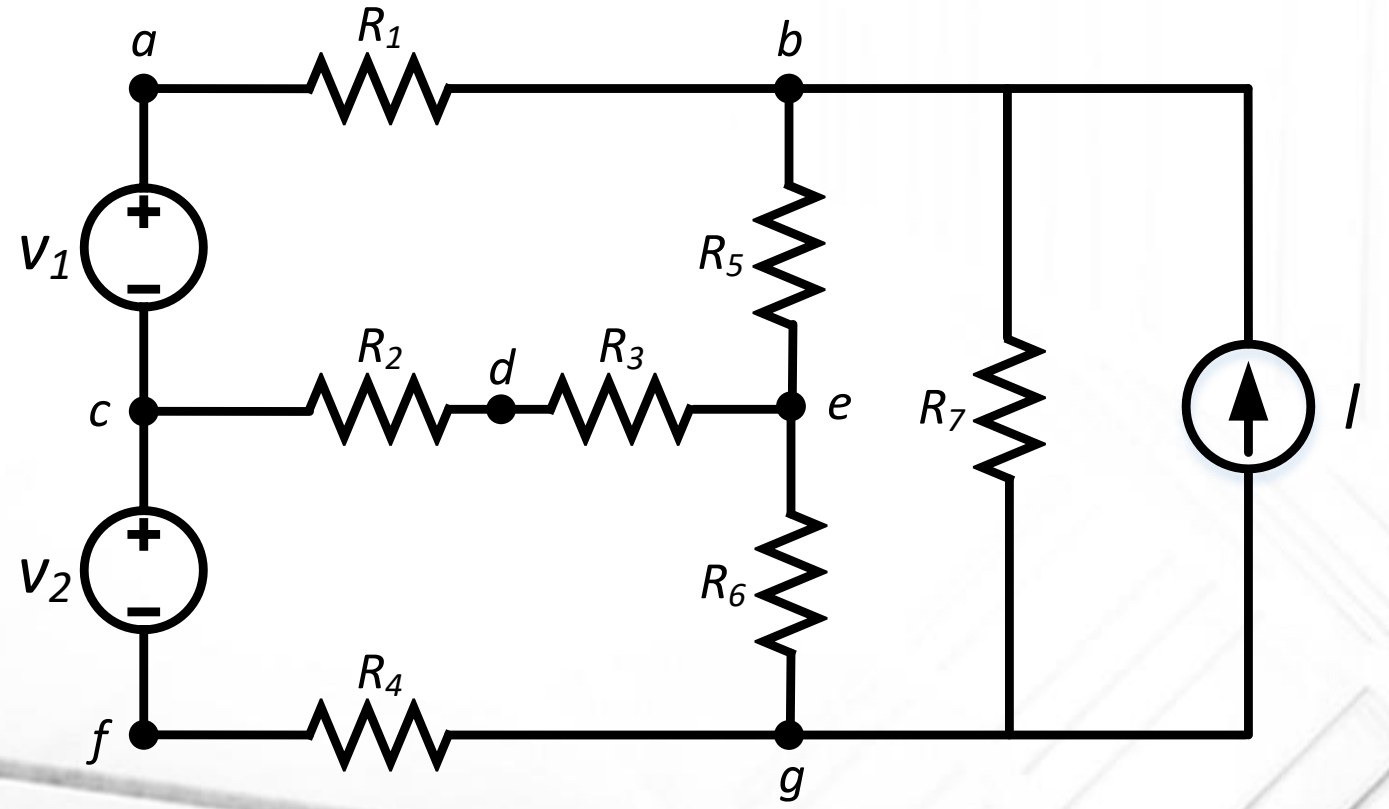


More Terminology

Name	Definition
Node	A point where two or more circuit elements meet
Essential Node	A point where three or more circuit elements meet
Path	A trace that adjoins basic elements, with no elements included more than once
Branch	A path that connects two nodes
Essential Branch	A path that connects two essential nodes without passing through an essential node
Loop	A path whose last node is the same as the starting node
Mesh	A loop that does not enclose any other loops

Example

Find all nodes, essential nodes, branches, essential branches, meshes, two paths that are not loops or essential branches, two loops that are not meshes.



Systematic Approach

Let's talk about how many equations are needed to solve a circuit!

If ***b*** is the number of branches with unknown current, then we need ***b*** independent equations to solve for the currents.

Let ***n*** be the number of nodes.

Then, the ***b*** independent equations required can be found from:

- $n - 1$ KCL equations for $n - 1$ independent nodes
- $b - (n - 1)$ KVL equations for $b - (n-1)$ independent meshes

Systematic Approach

If $\boldsymbol{b_e}$ is the number of essential branches with unknown current, then we need $\boldsymbol{b_e}$ independent equations to solve for the currents.

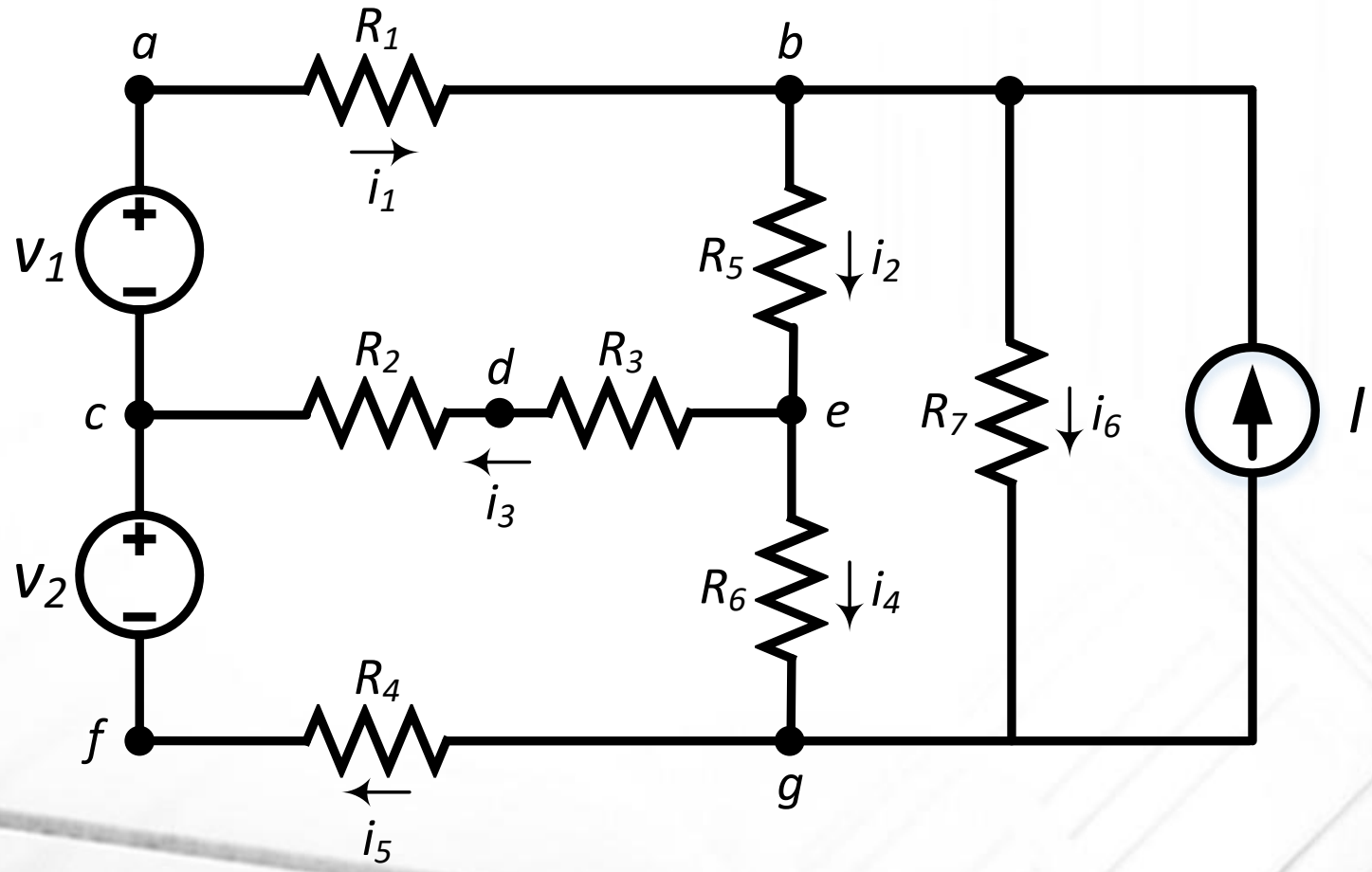
Let $\boldsymbol{n_e}$ be the number of essential nodes.

Then, the $\boldsymbol{b_e}$ independent equations required can be found from:

- $n_e - 1$ KCL equations for $n_e - 1$ independent essential nodes
- $b_e - (n_e - 1)$ KVL equations for $b_e - (n_e - 1)$ independent meshes

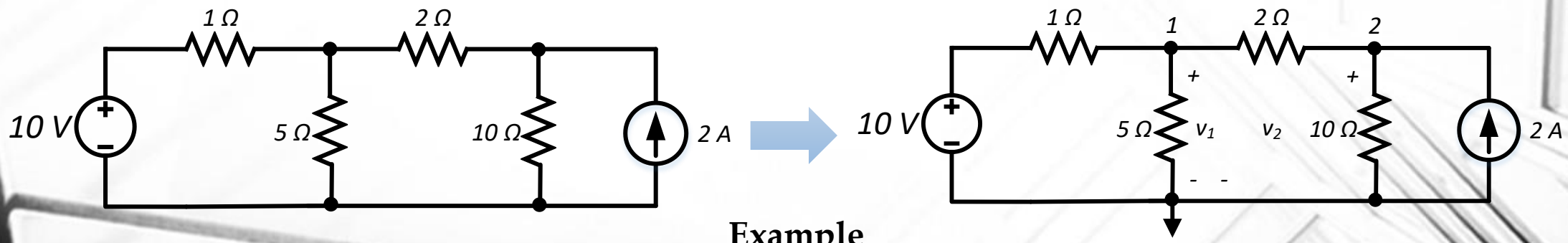
Example

Find the KCL and KVL equations for the following circuit:



Node-Voltage Method

1. Make sure the layout of the circuit is planar.
2. Mark the essential nodes of the circuit.
3. Select on of the essential nodes to be the "reference node". Usually the node with the most branches is selected.
4. Define the node voltages on your circuit diagram. A node voltage is defined as the voltage rise from the reference node to the non-reference node.



Example

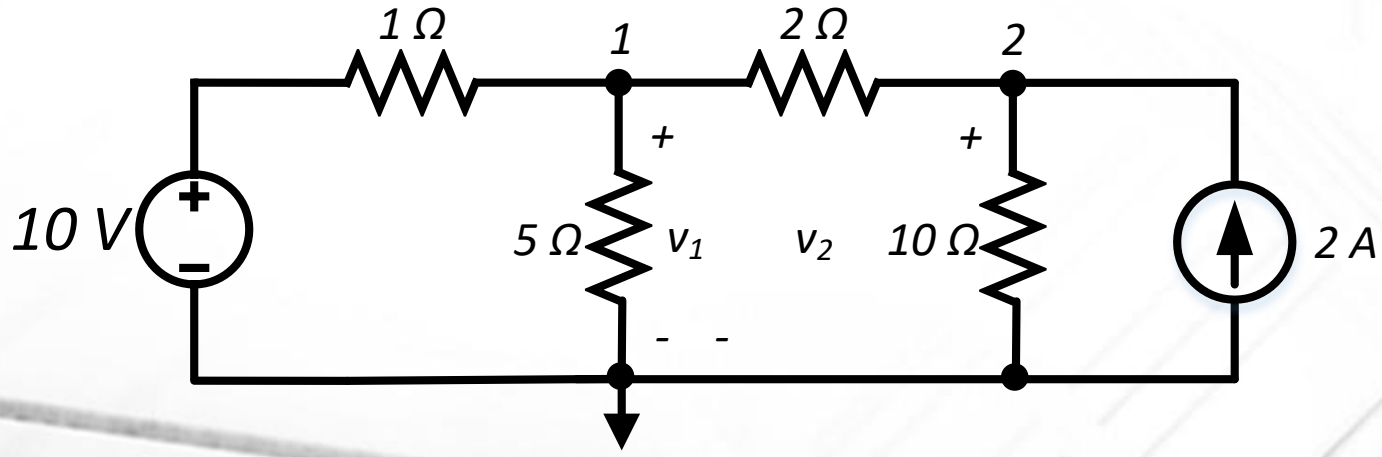
Node-Voltage Method

5. Find the Node-Voltage equations. We do this by finding the KCL equations for each non-reference node expressed as a function of the node voltages.

6. Solve for the non-reference node voltages.

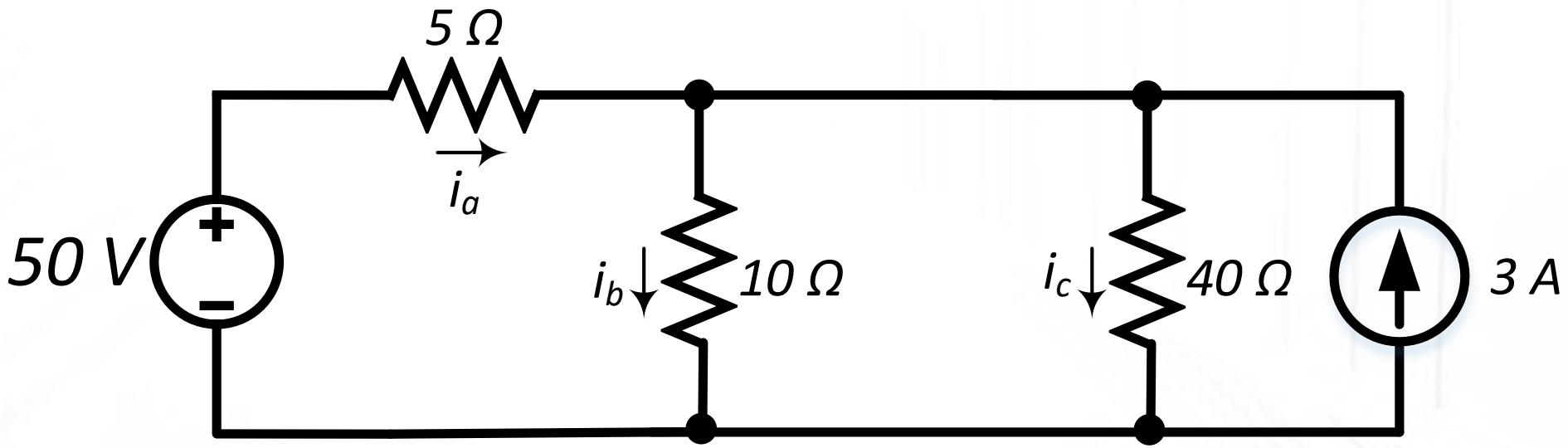
7. Find the rest of the unknown variables using the node voltages.

Example: Use the Node-Voltage Method on the following circuit.



Example

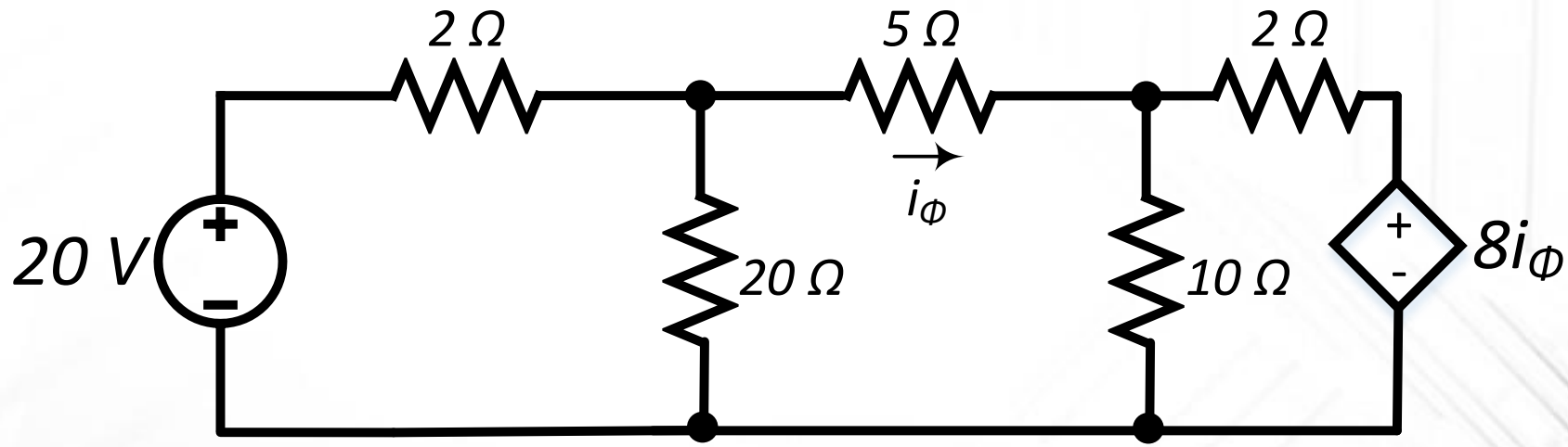
Using the Node Voltage Method, find the branch currents i_a, i_b, and i_c.



Node-Voltage Method: Dependent Sources

Equations defining dependent sources ("constraint equations") must be included in the node voltage equations.

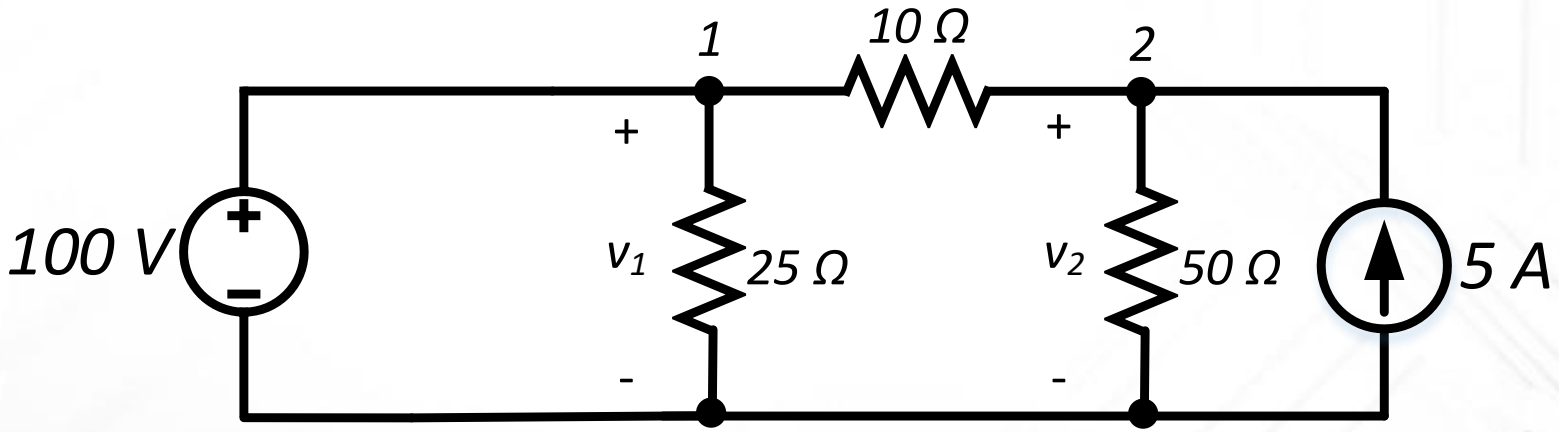
Example: Using the Node-Voltage Method, find the power dissipated in the 5Ω resistor.



Node-Voltage Method: Special Cases

Special Case #1: When a voltage source is between a reference node and a non-reference node, the voltage of the non-reference node is known (it has the same voltage as that of the voltage source). Thus, there is no need to write the Node Voltage equation for this non-reference node.

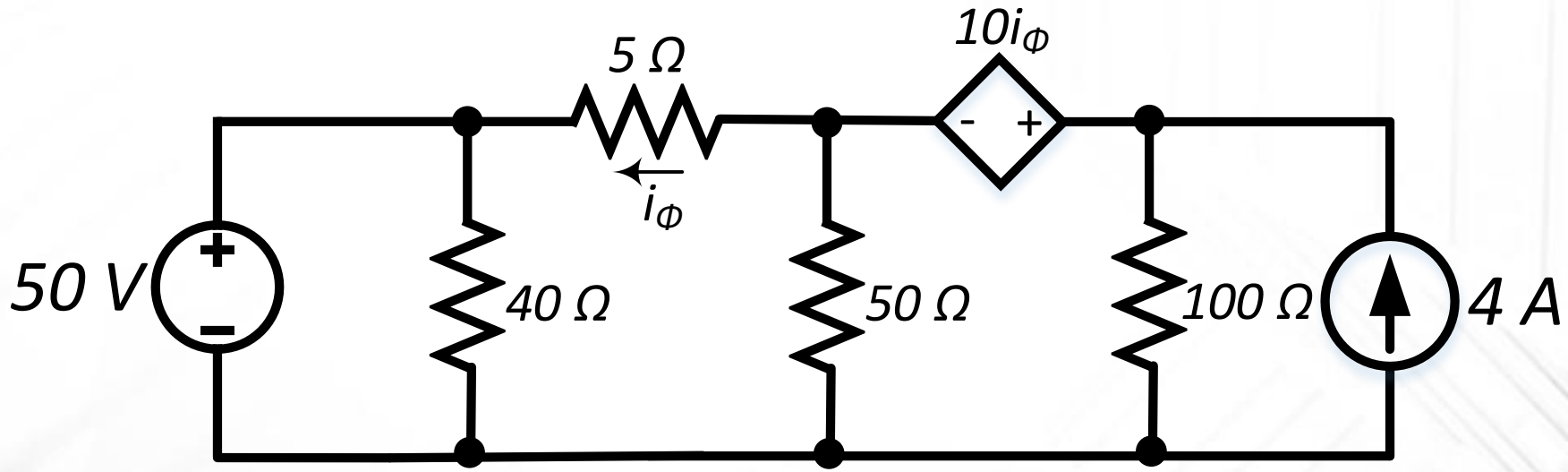
Example: Solve for v_1 and v_2.



Node-Voltage Method: Special Cases

Special Case #2: When a voltage source is between two non-reference nodes, the difference between the two node voltages is constrained to equal the voltage source.

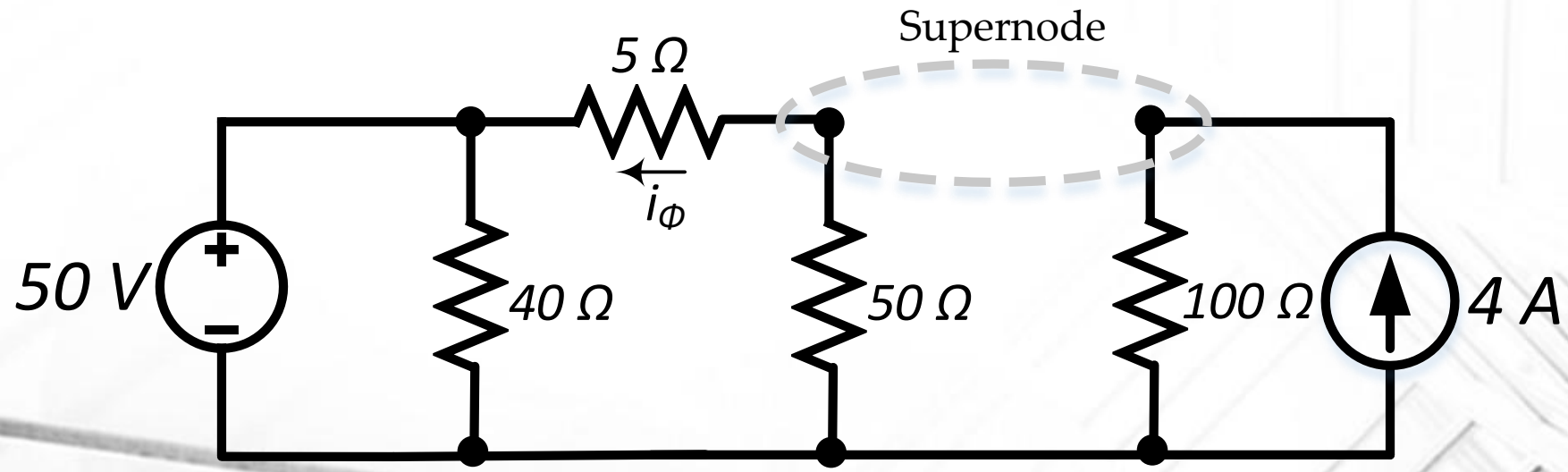
Example: Solve using the Node-Voltage Method.



Node-Voltage Method: Special Cases

Supernodes: When there is a voltage source is between two essential nodes, the two nodes can be combined to form a supernode. This supernode is considered as one node, because we know the relationship between the two nodes. This is a simpler way to approach Special Case #2.

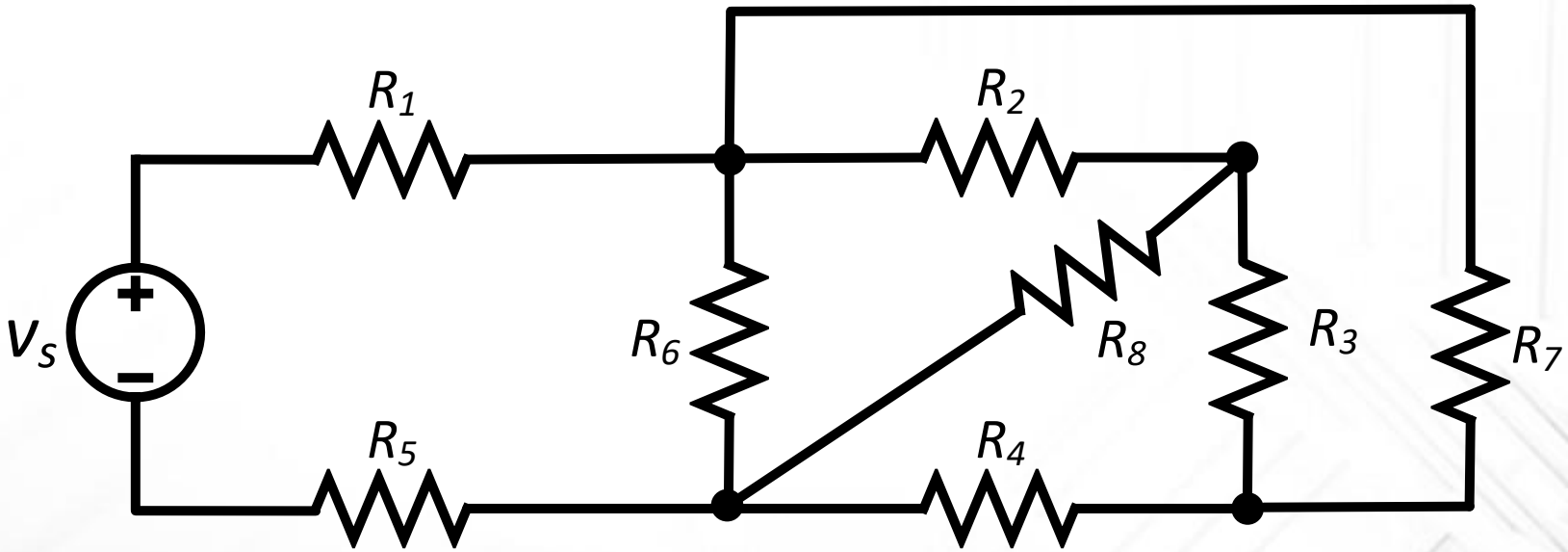
Example: Solve the previous example by using a supernode.



Mesh-Current Method

The Mesh-Current Method is a circuit analysis method that enables us to describe a circuit in terms of $b_e - (n_e - 1)$ equations, and is only applicable to planar circuits.

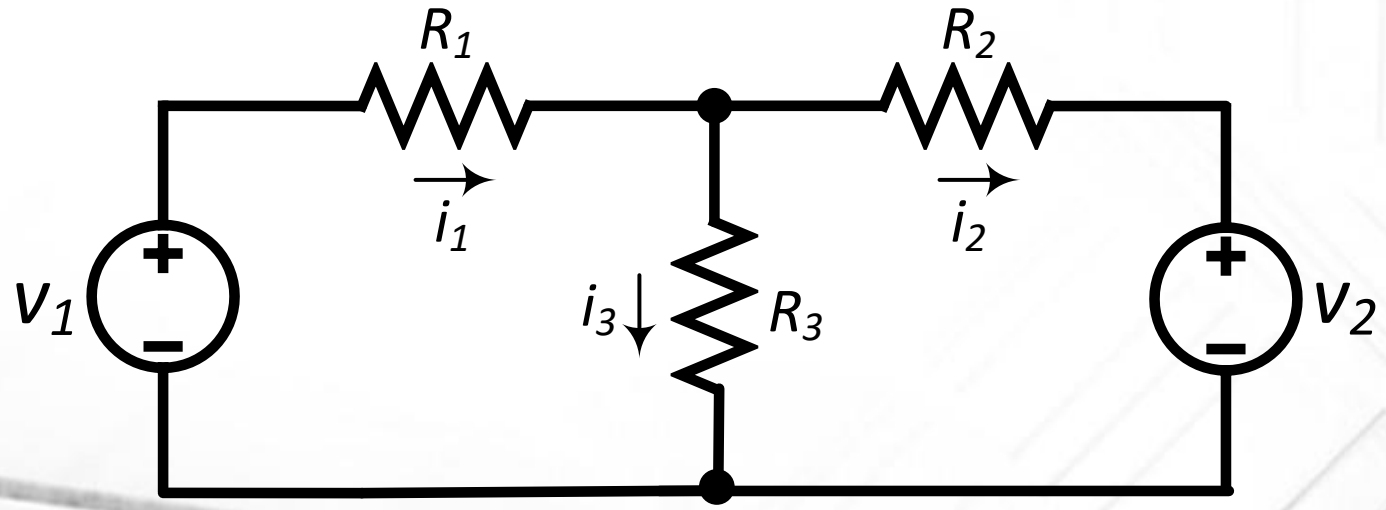
Example: Define the mesh currents.



Mesh-Current Method

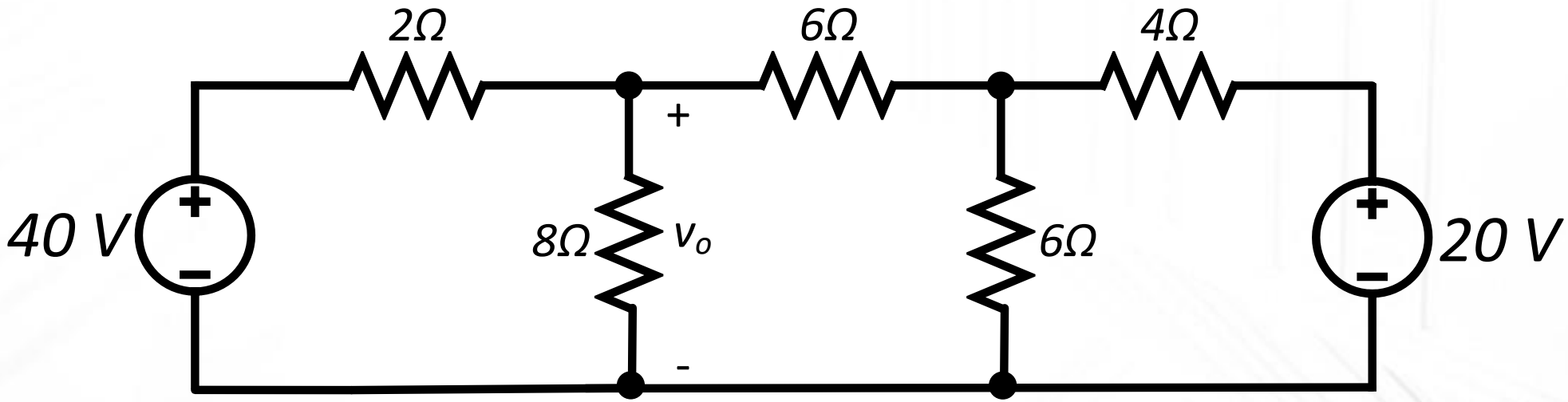
1. Identify $b_e - (n_e - 1)$ meshes.
2. Write the KVL equations for the meshes as a function of the mesh currents.
3. Solve for the currents.

Example: Use the Mesh-Current Method to analyze the following circuit.



Example

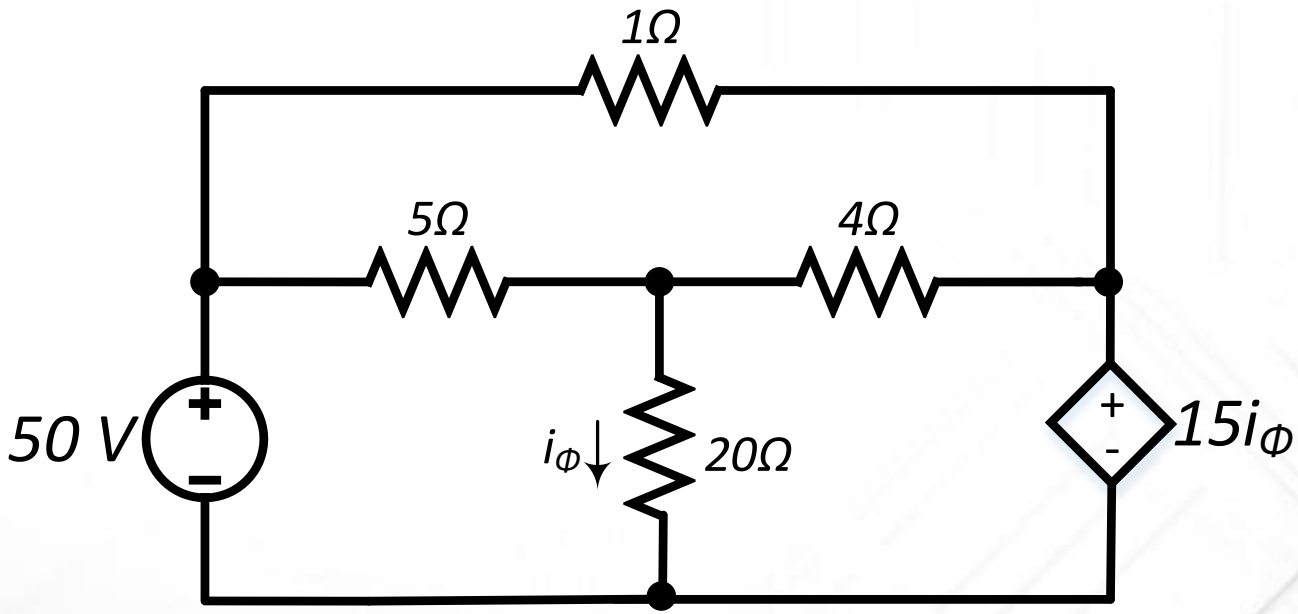
Use the Mesh-Current Method to (a) determine the power associated with each voltage source in the circuit, and (b) determine the voltage across the 8Ω resistor.



Mesh-Current Method: Dependent Sources

Equations defining dependent sources ("constraint equations") must be included in the mesh current equations.

Example: Use the Mesh-Current Method to determine the power dissipated in the 4Ω resistor.



Mesh-Current Method: Special Case

Special Case: When a branch contains a current source, we may reduce the number of unknown mesh currents.

Example: Use the mesh-current method on the following circuit.

